NBA Edge Model — Setup Guide

Every command, copy-paste ready. Accounts for your current machine state.

Your current state (from your terminal output):

- Homebrew: installed (5.1.3)
- Python 3.12: installed but PATH not updated (python3 still shows 3.9)
- Git: installed and upgraded to 2.53.0
- Git config: not set (the two commands were put on one line)
- Project zip: not downloaded yet (file not found in Downloads)

What this guide covers:

Phase 0: Fix your current issues and finish machine setup

Phase 1: Download project, set up environment, pull real NBA data

Phase 2: Run feature engineering, train models, backtest

Phase 3: Run improvements (calibration, tuning, tests)

Phase 4: Push to GitHub

IMPORTANT RULE: Run each command ONE AT A TIME. Do not paste multiple commands at once. Paste one line, press Enter, wait for it to finish, then paste the next line.

PHASE 0 — Fix Current Issues (~5 min)

Goal: Fix the three problems from your last terminal session.

Step 0.1 — Fix Python PATH

Python 3.12 is installed but your terminal still uses 3.9. Fix it:

Run this line:

```
echo 'export PATH="/opt/homebrew/opt/python@3.12/libexec/bin:$PATH"' &gt;&gt; ~/.zshrc
```

Then run this line:

```
source ~/.zshrc
```

Now verify:

```
python3 --version
```

Should say **Python 3.12.13** (not 3.9).

> *If it still says 3.9: close Terminal completely (Cmd+Q), reopen it, and try python3 --version again.*

Step 0.2 — Fix Git config

Last time both commands were on one line. Run them separately:

Run this line:

```
git config --global user.name "Omed Akbari"
```

Then run this line:

```
git config --global user.email "omedakbari@berkeley.edu"
```

Verify:

```
git config --global --list
```

Should show your name and email.

Step 0.3 — Find or re-download the zip file

The zip file did not land in Downloads last time. Let us find it or get it fresh.

First, check if it downloaded with a different name:

```
ls ~/Downloads/ | grep -i nba
```

If you see a file like **nba-edge-model-final.zip** or **nba_model_project.zip** or **nba-edge-model.zip**, note the exact name. You will use it in Step 1.2.

If you see nothing: go back to the Claude conversation where you got the files, click the zip file link again, and wait for it to download. Then run the ls command above again to confirm it appeared.

> *If the file keeps failing to download: ask the other LLM to help you build the project from scratch using the LLM Context Document (the other file you are getting from this conversation).*

CHECKPOINT

- [ ] python3 --version shows 3.12.x
- [ ] git config --global --list shows your name and email
- [ ] You can see the zip file name in your Downloads folder

PHASE 1 — Project Setup + Data Pull (~4 hours, mostly waiting)

Goal: Unzip the project, install packages, pull real NBA data.

Step 1.1 — Go to projects folder

```
cd ~/projects
```

(This folder already exists from your last session.)

Step 1.2 — Copy and unzip the file

Replace FILENAME below with the actual name you saw in Step 0.3:

```
cp ~/Downloads/FILENAME.zip ~/projects/
```

For example, if the file is called nba-edge-model-final.zip:

```
cp ~/Downloads/nba-edge-model-final.zip ~/projects/
```

Now unzip it:

```
cd ~/projects
```

```
unzip *.zip
```

If it created a folder called nba_model_project, rename it:

```
mv nba_model_project nba-edge-model
```

If it created a folder already called nba-edge-model, skip the rename. Go into it:

```
cd nba-edge-model
```

Verify:

```
ls
```

You should see files like: ARCHITECTURE.md, README.md, config, data, requirements.txt, scripts, src, tests

> *If ls shows nothing or an error: the unzip may have created a differently named folder. Run: ls ~/projects/ to see what folders exist, then cd into the right one.*

Step 1.3 — Create virtual environment

```
python3 -m venv venv
```

```
source venv/bin/activate
```

You should see **(venv)** at the start of your terminal prompt. This is critical.

> *If you do NOT see (venv), something went wrong. Try: python3.12 -m venv venv and then source venv/bin/activate*

Step 1.4 — Install packages

```
pip install --upgrade pip
```

Then run this single long line (it is one command):

```
pip install nba_api pandas numpy scikit-learn xgboost lightgbm scipy pyarrow matplotlib seaborn
tqdm jupyter streamlit requests python-dotenv
```

This takes 2-3 minutes. Wait for it to finish.

> *If you get errors about 'command not found' for pip: make sure (venv) is showing. If not, run: source venv/bin/activate*

Step 1.5 — Set up the shortcut

```
echo 'alias nba="cd ~/projects/nba-edge-model &amp;&amp; source venv/bin/activate"' &gt;&gt;
~/.zshrc
```

```
source ~/.zshrc
```

Now any time you open a new Terminal window, just type **nba** to jump into the project.

Step 1.6 — Initialize database

Make sure you are in the project folder with (venv) active, then:

```
python -c "from src.utils.db import init_db; init_db()"
```

Should print: Database initialized

Step 1.7 — Test the pipeline

```
python scripts/pull_season_data.py --test
```

Should print Jokic's stats and a test complete message.

> *If you get 'ModuleNotFoundError': you are not in the venv. Run: source venv/bin/activate*
>
> *If you get a connection/timeout error: wait 1 minute and try again. NBA.com rate-limits requests.*

Step 1.8 — Pull the current season (30-60 min wait)

```
python scripts/pull_season_data.py --season 2024-25
```

You will see a progress bar. **Let it run.** Go do something else.

When it finishes, verify:

```
python3 -c "from src.utils.db import get_connection; import pandas as pd; conn=get_connection();
print(pd.read_sql('SELECT COUNT(*) as rows FROM game_logs', conn))"
```

Step 1.9 — Pull all historical seasons (3-5 hour wait)

Before running this, prevent your Mac from sleeping:

Go to System Settings, then Displays, then Advanced, and turn off auto-sleep.

```
python scripts/pull_season_data.py --backfill
```

This takes 3-5 hours. Leave your laptop open and plugged in.

> *If it crashes partway through: just run the exact same command again. It picks up where it left off.*

When it finishes, verify all seasons are present:

```
python3 -c " from src.utils.db import get_connection import pandas as pd conn = get_connection()
print(pd.read_sql('SELECT season, COUNT(*) as games FROM game_logs GROUP BY season ORDER BY
season', conn).to_string(index=False)) "
```

CHECKPOINT

- [ ] (venv) shows in your terminal prompt
- [ ] python scripts/pull_season_data.py --test works
- [ ] Current season data is in the database
- [ ] All 6 seasons show data after backfill completes

PHASE 2 — Features, Models, Backtest (~15 min)

Goal: Transform raw data into features, train models, run backtests.

Make sure you are in the project with (venv) active. If you opened a new Terminal, type: **nba**

Step 2.1 — Run feature engineering

```
python -m src.features.pipeline
```

Takes 1-5 minutes. Should end with 'Final dataset: 150,000+ rows'

> *If you get an import error: make sure you are in ~/projects/nba-edge-model with (venv) active.*

Step 2.2 — Train the models

```
python -m src.models.player_props
```

Takes 2-10 minutes. You will see MAE scores for each stat (pts, reb, ast, fg3m, pra, reb_ast).

Each model should beat the L10 baseline — you will see 'Improvement: X%' for each one.

Step 2.3 — Run the backtest

```
python -m src.models.edge_finder
```

Prints hit rates at different confidence thresholds (55%, 60%, 65%, 70%, 75%).

Key thing to look for: at 70% threshold, does the actual hit rate land near 65-75%?

Step 2.4 — Run the unit tests

```
python tests/test_pipeline.py
```

Should show tests passing. Some tests may fail if they reference synthetic model files — that is OK. The database and feature tests should all pass.

CHECKPOINT

- [ ] data/processed/features.parquet (or features.csv) exists
- [ ] Model files exist in data/processed/models/
- [ ] Backtest report printed without errors
- [ ] Most tests pass

PHASE 3 — Push to GitHub (~10 min)

Step 3.1 — Initialize git

```
cd ~/projects/nba-edge-model
```

```
git init
```

```
git add .
```

```
git commit -m "NBA edge model - full pipeline with real data"
```

Step 3.2 — Create repo on GitHub

1. Open your browser and go to: github.com/new
2. Repository name: **nba-edge-model**
3. Set to **Private**
4. Do NOT check any boxes (no README, no gitignore, no license)
5. Click **Create repository**

Step 3.3 — Push your code

Replace YOUR_USERNAME with your actual GitHub username:

```
git remote add origin https://github.com/YOUR_USERNAME/nba-edge-model.git
```

```
git branch -M main
```

```
git push -u origin main
```

It will ask for your username and password.

For the password: you need a Personal Access Token, not your actual password.

1. Go to: github.com/settings/tokens
2. Click 'Generate new token (classic)'
3. Give it a name like 'nba-model'
4. Check the 'repo' checkbox
5. Click 'Generate token' at the bottom
6. Copy the token (starts with ghp_)
7. Paste it as your password when Terminal asks

CHECKPOINT

[ ] Your repo shows files on github.com

TROUBLESHOOTING — Common Issues

'command not found: python3'

Close Terminal (Cmd+Q), reopen it, try again. If still broken:

```
brew link python@3.12 --force
```

```
source ~/.zshrc
```

'No module named X'

You are not in the virtual environment. Run:

```
cd ~/projects/nba-edge-model
```

```
source venv/bin/activate
```

Then try your command again.

'Connection error' or 'timeout' during data pull

NBA.com rate-limits. Wait 2 minutes and run the same command again.

Data pull crashes partway through

Run the exact same command again. It skips what it already pulled.

'Permission denied' errors

Add sudo in front of the command. It will ask for your Mac password.

pip install fails

Make sure (venv) is showing in your prompt. If not:

```
source venv/bin/activate
```

Then try pip install again.

python3 --version still shows 3.9 after the PATH fix

Try running the specific version directly:

```
python3.12 --version
```

If that works, use python3.12 everywhere this guide says python3.

Also, when creating the venv: python3.12 -m venv venv

Cannot find downloaded zip file

Open Finder on your Mac. Press Cmd+Shift+G. Type ~/Downloads and press Enter. Look for any file with 'nba' in the name. If nothing: the download failed. Go back to the Claude chat and click the file again.

git push asks for password and rejects it

You need a Personal Access Token, not your GitHub password. See Step 3.3 above.

For anything else

Copy the EXACT error message from Terminal. Paste it to the other LLM along with the context document. It will know what to do.